Experiment 1

Aim:

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures.

Objectives of this Experiment:

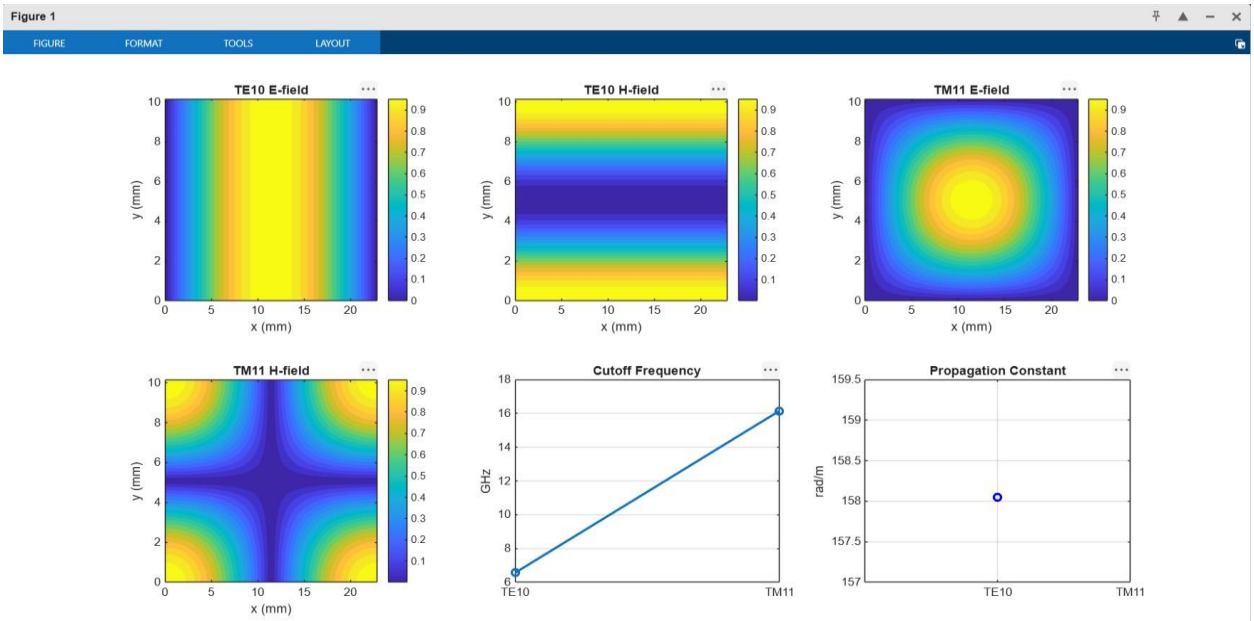
1. To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.
2. To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.
3. To determine the cutoff frequency, propagation constant and field characteristics of selected TE and TM modes.
4. To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.

Software Required: MATLAB

Code:

```
clc; clear; close all; c=3e8; a=22.86e-3; b=10.16e-3; f=10e9; fc10=c/(2a);
fc11=c/2sqrt((1/a)^2+(1/b)^2); k=2pif/c; beta10=ksqrt(1-(fc10/f)^2); x=linspace(0,a,80);
y=linspace(0,b,60); [X,Y]=meshgrid(x1e3,y1e3); E10=sin(piX/(a1e3));
H10=0.5(1+cos(2piY/(b1e3))); E11=sin(piX/(a1e3)).sin(piY/(b1e3));
H11=abs(cos(piX/(a1e3)).cos(piY/(b*1e3))); figure('Color','w','Position',[100 50 1400 700]);
subplot(2,3,1); contourf(X,Y,E10,20,'LineColor','none'); title('TE10 E-field'); xlabel('x (mm)');
ylabel('y (mm)'); colorbar; subplot(2,3,2); contourf(X,Y,H10,20,'LineColor','none'); title('TE10
H-field'); xlabel('x (mm)'); ylabel('y (mm)'); colorbar; subplot(2,3,3);
contourf(X,Y,E11,20,'LineColor','none'); title('TM11 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
colorbar; subplot(2,3,4); contourf(X,Y,H11,20,'LineColor','none'); title('TM11 H-field'); xlabel('x
(mm)'); ylabel('y (mm)'); colorbar; subplot(2,3,5); plot([1 2],[fc10 fc11]/1e9,'-o','LineWidth',2);
set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); title('Cutoff Frequency'); ylabel('GHz');
grid on; subplot(2,3,6); plot(1,beta10,'bo','LineWidth',2); set(gca,'XTick',[1
2],'XTickLabel',{'TE10','TM11'}); title('Propagation Constant'); ylabel('rad/m'); ylim([157
159.5]); grid on; fprintf('TE10 cutoff = %.3f GHz\n',fc10/1e9); fprintf('TM11 cutoff = %.3f
GHz\n',fc11/1e9); fprintf('TE10 beta = %.3f rad/m\n',beta10); fprintf('TE10 propagates; TM11 is
below cutoff at 10 GHz.\n');
```

Output:



TE10 cutoff = 6.562 GHz

TM11 cutoff = 16.156 GHz

TE10 beta = 158.046 rad/m

TE10 propagates; TM11 is below cutoff at 10 GHz.

Exp 1

Theoritical value of Exp 1

1) TE_{10} mod cutoff frequency

$$f_{c10} = \frac{c}{2a}$$

$$f_{c10} = \frac{3\times10^8}{2(0.02286)}$$

$$f_{c10} = 6.562\ GHz$$

TM_{11} mode cutoff freq

$$f_{c(mn)} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$$

m=1, n=1

$$f_{c11} = \frac{3\times10^8}{2}\sqrt{\left(\frac{1}{0.02286}\right)^2+\left(\frac{1}{0.01016}\right)^2}$$

$$f_{c11} = 16.156\ GHz$$

TE_{10} propagation constant

$$\beta = \frac{2\pi f}{c}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$$\beta = \frac{2\pi(10^{10})}{3\times10^8}\sqrt{1-\left(\frac{6.562}{10}\right)^2}$$

$$\beta = 158.046\ rad/m$$

TM_{11} attenuation constant

$$\alpha = k_0\sqrt{\left(\frac{f_c}{f}\right)^2-1}$$

$k_0 = 2\pi f/c$

$$\alpha = \frac{2\pi(10^{10})}{3\times10^8}\sqrt{\left(\frac{16.156}{10}\right)^2-1}$$

$$\alpha = 265.769\ Np/m$$

TE_{10} mode exist

TM_{11} mode does not propagate